head
head

1. **Answer the question below by choosing the correct response.**

Which of the following is the expanded form of 4,307.67?

- $(4 \times 1{,}000) + (3 \times 100) + (7 \times 1) + \left(6 \times \frac{1}{100}\right) + \left(7 \times \frac{1}{10}\right)$
- $(4 \times 1{,}000) + (3 \times 100) + (7 \times 1) + \left(6 \times \frac{1}{10}\right) + \left(7 \times \frac{1}{100}\right)$
- $(4 \times 1{,}000) + (3 \times 100) + (7 \times 10) + \left(6 \times \frac{1}{10}\right) + \left(7 \times \frac{1}{100}\right)$
- $(4 \times 1{,}000) + (3 \times 100) + (7 \times 10) + (6 \times 1) + \left(7 \times \frac{1}{100}\right)$

2. Answer the question below by choosing the correct response.

Hugh needs to fill milk jugs to take to sell to a grocery market. The number of jugs, n, needed to fill the order is $n = 3w + 2$, where w is the number of weeks until the next delivery.

Which of the following statements is true about the variables n and w?

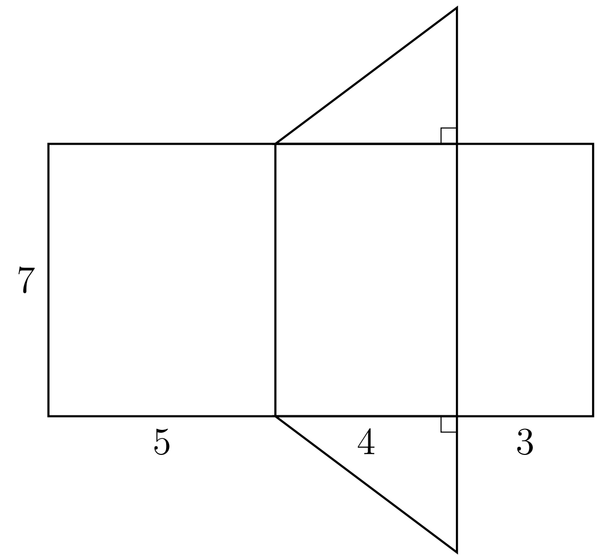
- ◯ n is the independent variable, and w is the dependent variable.
- ◯ w is the independent variable, and n is the dependent variable.
- ◯ Both n and w are dependent variables.
- ◯ Both n and w are independent variables.

3. Answer the question below by choosing the correct response.

Tiana has T tiaras, and Cinderella has C tiaras. If Tiana has three times as many tiaras as Cinderella and together they have a total of 40 tiaras, which of the following proportions can be used to find out how many tiaras Tiana has?

- ◯ $T : C = 3 : 4$
- ◯ $C : T = 3 : 4$
- ◯ $T : (C + T) = 3 : 4$
- ◯ $C : (C + T) = 3 : 4$

4. Answer the question below by choosing the correct response.



The net shown above forms a right triangular prism. Which of the following is the surface area of the right triangular prism in square units?

- 84
- 96
- 108
- 120

5. Answer the question below by choosing the correct response.

Item	Cost per Package	
Pens	\$2.50	4
Pencils	\$2.00	6
Staples	\$2.00	2
Notebooks	\$3.50	2

The table above shows the costs of office supplies. Andy buys 4 packages of pens, 6 packages of pencils, 2 boxes of staples, and 2 boxes of notebooks. Andy also has to pay 5% sales tax. How much will the supplies cost?

- ◯ \$10.00
- ◯ \$10.50
- ◯ \$33.00
- ◯ \$34.65

6. Answer the question below by choosing the correct response.

A box contains B books and M magazines, and there are three fewer books than twice the number of magazines.

Which of the following equations can be used to represent algebraically the relationship between the number of books and the number of magazines in the box?

- ⬭ $B = 3 - 2M$
- ⬭ $B - 3 = 2M$
- ⬭ $B + 3 = 2M$
- ⬭ $2A = M - 3$

7. Answer the question below by choosing the correct response.

$$20A + 15P = 750$$

In a certain month, a farmer earned a total of \$750 selling boxes of pears and apples. She sells each box of pears for \$15 and each box of apples for \$20. In the formula above, A represents the number of boxes of apples the farmer sold and P represents the number of boxes of pears sold by the farmer this month.

If the farmer sold 15 boxes of apples that month, how many boxes of pears did she sell?

- ◯ 10
- ◯ 20
- ◯ 30
- ◯ 70

8. Answer the question below by choosing the correct response.

A regular octagon can be decomposed into how many regular triangles?

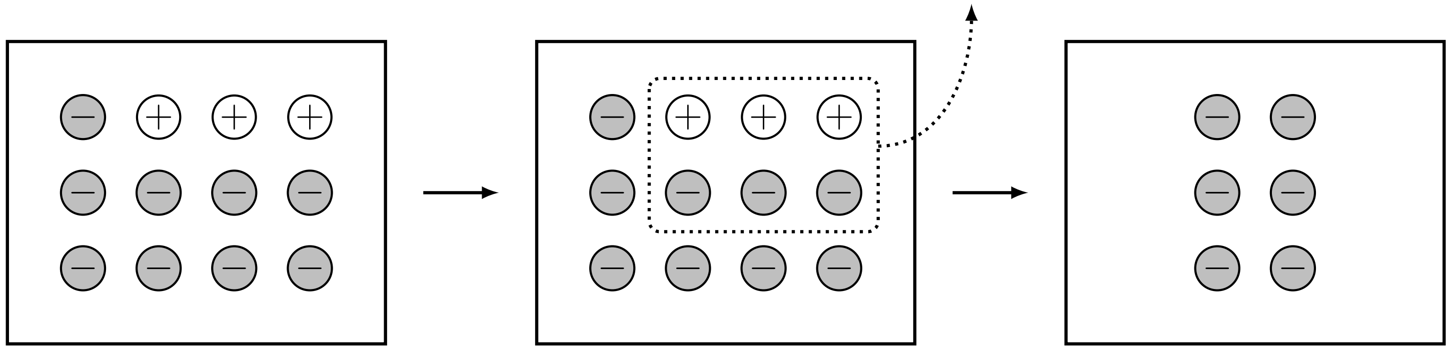
- ◯ 4
- ◯ 5
- ◯ 6
- ◯ 8

9. Answer the question below by choosing the correct response.

What is the prime factorization of 360?

- $3^2 \times 8 \times 5$
- $2^3 \times 5 \times 9$
- $2^3 \times 3^2 \times 5$
- $4 \times 9 \times 10$

10. Answer the question below by choosing the correct response.



Which of the following addition sentences represent the model shown?

- $-9x + 3x = -6x$
- $-12x + 6x = -6x$
- $12x - (-6x) = -6x$
- $-4x + (-2x) = -6x$

11. Indicate all of your answers.

$$12, 14, 15, 16, 19, 20, 20, 20, 22, 23, 23, 25, 31, 31, 32, 33, 35$$

The list shows the ages of people at Mrs. Bings' birthday party. Which of the following statements about the ages must be true?

- ☐ The mode is 22 years.
- ☐ The median is 22 years.
- ☐ The mean is 23 years.
- ☐ The range is 42 years.

12. Answer the question below by choosing the correct response.

Which of the following is an algebraic equation?

- ◯ $\dfrac{\sqrt{2x} + 3y}{5}$
- ◯ $\sqrt{2}x - 3 = 5$
- ◯ $2(3 - 2) = 5 \div (3 + \sqrt{2})$
- ◯ $5 + 2 - \dfrac{10}{3}$

13. Answer the question below by choosing the correct response.

Which of the following is equivalent to $\frac{1}{3} \div 9$

- ⬭ $\frac{1}{3} \times \frac{9}{1}$
- ⬭ $\frac{1}{3} \times \frac{1}{9}$
- ⬭ $\frac{3}{1} \times \frac{9}{1}$
- ⬭ $\frac{3}{1} \times \frac{1}{9}$

14. Answer the question below by choosing the correct response.

Term	**Figure(s)**	**Lines**
1	⬠	5
2	⬠ + ⬠	12
3	⬠ + ⬠ + ⬠	19
⋮	⋮	⋮

In the sequence of figures shown in the table above, the first term is made of a single shape and each successive term is made by adding one shape as indicated by the pattern. 5 line segments are needed to form a pentagon, and 2 line segments are needed to form each cross. Which of the following expression can be used to find the number of line segments needed to form the n^{th} term in each sequence of figures?

- ⬭ $n + 7$
- ⬭ $7n - 2$
- ⬭ $n^2 + 3$
- ⬭ $(n + 1)^2 - 4$

15. Answer the question below by choosing the correct response.

The digit 1 is in what place in the number 6.283185?

- Hundredths place
- Thousandths place
- Ten-thousandths place
- Hundred-thousandths place

16. Answer the question below by choosing the correct response.

Roger the rabbit left his rabbit hole at 2:55 P.M., and it took him $\frac{2}{3}$ of an hour to get to the forest. At what time did Roger get to the forest?

- 3:25 P.M.
- 3:35 P.M.
- 3:45 P.M.
- 3:55 P.M.

17. Answer the question below by choosing the correct response.

Harry the magician agreed to pay his agent 30% of the ticket sales from his shows. If the ticket sales from Harry's last show was $201.30, how much does Harry owe his agent?

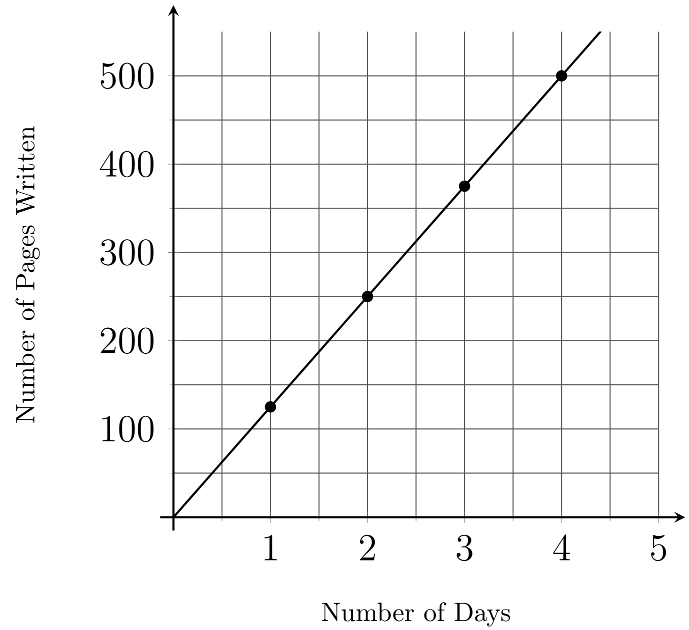
- $6.04
- $60.39
- $67.10
- $603.9

18. Answer the question below by choosing the correct response.

In the xy-plane, point R has coordinates $(3,2)$ and point S has coordinates $(8,2)$. Which of the following could be the coordinates for point T so that the area of triangle RST is equal to 15 square units?

- ⬭ $(6,8)$
- ⬭ $(6,4)$
- ⬭ $(8,4)$
- ⬭ $(3,6)$

19. Answer the question below by choosing the correct response.



Stephen Koontz is writing a 750-page novel at a constant rate. The graph shows the number of pages, y, he has written after x days of writing. Assuming he continues writing as the same rate, how many days will it him to finish writing the novel?

- 4
- 6
- 8
- 10

20. Indicate all of your answers.

$$2, 4, 8, \ldots$$

The first three terms of a sequence are shown. Which of the following mathematical relationships could describe the terms of the sequence?

Select **all** that apply.

- ☐ The first term is 2. Each subsequent term of the sequence is obtained by multiplying a constant by the preceding term.
- ☐ The first term is 2. Each subsequent term of the sequence is obtained by adding a constant to the preceding term.
- ☐ The first term is 2. Each subsequent term of the sequence is obtained by squaring the previous term.
- ☐ The first term is 2. Each subsequent term of the sequence is obtained by adding the preceding term and a quantity that increases by 2 with each new term.
- ☐ The first term is 2. Each subsequent term of the sequence is obtained by squaring the previous term and subtracting a constant that increases by 8 with each new term.

21. Answer the question below by choosing the correct response.

x and y are digits in some number, and x represents 10^3times what y represents. Which of the following could be the place of x and y?

- ◯ x is the thousandths place, and y is the ones place.
- ◯ x is the tenths place, and y is the hundreds place.
- ◯ x is the tens place, and y is in the thousandths place.
- ◯ x is in the tens place, and y is in the hundredths place.

22. Answer the question below by choosing the correct response.

At a local high-school with 30 paid employees, one employee has an annual salary of \$135 thousand, three employees have an annual salary of \$50 thousand each, and the remaining have annual salaries of \$25 thousand each. Which of the following best represents a typical employee's annual salary?

- ◯ Mean
- ◯ Median
- ◯ Mid-range
- ◯ Range

23. Answer the question below by choosing the correct response.

$$(3z - 2z + 1) - (-7z + 8z - 1)$$

Which of the following is equivalent to the expression shown?

- ⬭ 2
- ⬭ 0
- ⬭ $2z + 2$
- ⬭ $2z$

24. Answer the question below by choosing the correct response.

1,364,520.79

In the number shown above, the value represented by the digit 4 is $4 \times c$, where c is a constant. Which of the following equals c?

- ⬭ 10^2
- ⬭ 10^3
- ⬭ 10^4
- ⬭ 10^5

25. Answer the question below by choosing the correct response.

Which of the following statements must be true about the two non-obtuse angles of an obtuse triangle?

- Both angles are obtuse.
- One angle is acute and one is angle is obtuse.
- Both angles are acute.
- The angles are congruent.

26. Answer the question below by choosing the correct response.

$$-5(x-2) \leq 20$$

Which of the following inequalities is equivalent to the inequality shown?

- ◯ $x \leq -2$
- ◯ $x \leq 2$
- ◯ $x \geq -2$
- ◯ $x \geq 2$

27. Answer the question below by choosing the correct response.

$$6 + 18 \div 2 \times (5 - 2)$$

Which of the following is equivalent to the expression shown?

- ◯ 4
- ◯ 9
- ◯ 33
- ◯ 36

28. Answer the question below by choosing the correct response.

Which list of numbers is ordered from least to greatest?

- ⬭ $\frac{13}{22}$, 0.44, $\frac{3}{5}$, $0.4\bar{4}$
- ⬭ 0.44, $0.4\bar{4}$, $\frac{13}{22}$, $\frac{3}{5}$
- ⬭ $\frac{3}{5}$, $\frac{13}{22}$, 0.44, $0.4\bar{4}$,
- ⬭ $0.4\bar{4}$, 0.44, $\frac{13}{22}$, $\frac{3}{5}$

29. Write your answer in the provided form.

A container in the shape of a cube is completely filled with 64 number cubes. Each number cube has an edge that is $\frac{3}{4}$ of a foot long. Find the volume, in cubic feet, of the container.

30. Answer the question below by choosing the correct response.

A banker evaluates the data in spreadsheet with records of total investments each day for one month. The banker deletes the greatest and smallest investment for the month from the spreadsheet. How is the data set most likely affected by removing this action?

- ◯ The mean and median of the data set will change, but the range will stay the same.
- ◯ The median and range of the data set will change, but the mean will stay the same.
- ◯ The mean and range of the data set will change, but the median will stay the same.
- ◯ The median, mean, and range of the data set will all change.

31. Answer the question below by choosing the correct response.

Determine which of the following two numbers estimates the number 1,409.147351 to the nearest tenth and to the nearest ten-thousandths, respectively?

- 1,409.1 and 1,409.1473
- 1,409.1 and 1,409.1474
- 1,409.2 and 1,409.1473
- 1,409.2 and 1,409.1474

32. Answer the question below by choosing the correct response.

James bought a new kayak with a price tag of \$899.99. He also has to pay sales tax rate at a rate 0f 6.9%. Determine which of the following proportions can be used to find the amount of sales tax, t, in dollars, for the new kayak?

- $\dfrac{6.9}{899.99} = \dfrac{t}{100}$
- $\dfrac{6.9}{899.99} = \dfrac{100}{t}$
- $\dfrac{6.9}{100} = \dfrac{t}{899.99}$
- $\dfrac{6.9}{100} = \dfrac{899.99}{t}$

33. Answer the question below by choosing the correct response.

$$(x+1)(y-2)+(y+2)(x+1)$$

Which of the following expressions is equivalent to the expression shown?

- $2y$
- $(x+1)$
- $2(x+1)$
- $2y(x+1)$

34. Answer the question below by choosing the correct response.

Tess made 453,592 milligrams of pasta. Which of the following represents the amount of pasta, in grams, that Tess made?

- ⬭ 45.3592
- ⬭ 453.592
- ⬭ 4535.92
- ⬭ 453,592,000

35. Answer the question below by choosing the correct response.

$3\frac{3}{4}\%$ is equivalent to which of the following numbers?

- 3.75
- 0.375
- 0.0375
- 0.00375

36. Indicate all of your answers.

$$5x^2 + 7y^2 - 13xy^2 + 4x^2y + 2$$

Which THREE of the following statements are correct about the given algebraic expression?

- ☐ The coefficients are 5, 7, -13, and 4
- ☐ The variables are x and y.
- ☐ There are like terms.
- ☐ There are five terms.
- ☐ The degree of the expression is 3.

37. Answer the question below by choosing the correct response.

Label	Probability
1	$\frac{1}{4}$
2	$\frac{3}{8}$
3	x
4	$\frac{1}{16}$

A spinner is divided into 16 congruent sections. Each section is labeled as either 1, 2, 3, or 4. The table shows the theoretical probability of landing on a number when spinning the spinner one time. How many sections are labeled 3?

- 2
- 4
- 5
- 11

38. Answer the question below by choosing the correct response.

Square A has an area of 1 square meter. The length of each side of square B is 1 meter greater than the length of each side of square A. The area of square B is how many times the area of square A?

- ◯ 1 times
- ◯ 2 times
- ◯ 4 times
- ◯ 8 times

39. Answer the question below by choosing the correct response.

Chef is making 4 chocolate cakes for a Kermit's birthday party. Each cake requires $2\frac{1}{4}$ cups of flour and $3\frac{1}{2}$ cups of chocolate syrup. Which of the following is the best estimate of the total amount of flour and chocolate syrup necessary to make the cakes?

- ◯ Less than 20
- ◯ Between 20 and 24 cups.
- ◯ Between 24 and 27
- ◯ More than 27

40. Answer the question below by choosing the correct response.

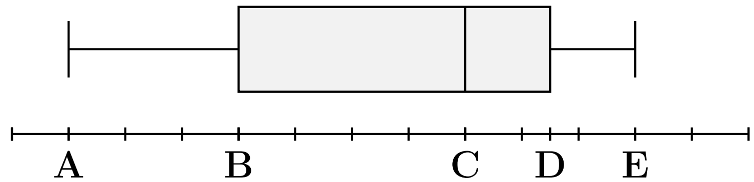
$$-d + 5r$$

What is the value of the expression shown when $d = 2$ and $r = -3$?

- ⬭ -17
- ⬭ -13
- ⬭ 13
- ⬭ 17

41. Answer the question below by choosing the correct response.

$$\{7,\ 9,\ -1,\ 2,\ 5,\ 10,\ 8,\ 2,\ 7,\ 0,\ 3,\ 7,\ 6\}$$



Note: Figure not drawn to scale.

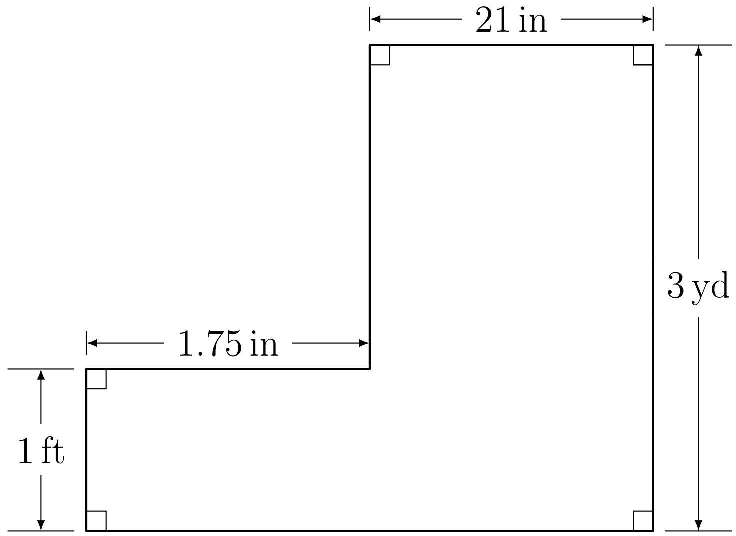
The list above shows the number of days missed this year by each of 13 students in a class. The data is summarized in the provided boxplot. What is the value of B on the boxplot?

- 2
- 3
- 5
- 6

42. Alisha's cat eats $\frac{1}{2}$ of a 10 pound bag of cat food every day. How many 10 pound bags must she buy to feed her cat for one week?

- 3
- 3.5
- 4
- 4.5

43. Answer the question below by choosing the correct response.



Find the area of the shown polygon.

- 25.75 square inches
- 165.5 square inches
- 777 square inches
- 2289 square inches

44. Answer the question below by choosing the correct response.

$$17x^3 + 2y^2 + 5x + 17y + 30x^3 - 14y^3$$

Which of the following shows all like terms of the expression above?

- $2y^2$, $17y$, and $-14y^3$
- $17x^3$ and $17y$
- $17x^3$, $30x^3$, and $-14y^3$
- $17x^3$ and $30x^3$

45. Answer the question below by choosing the correct response.

Job	Time
Cutting Hair	15 minutes
Tutoring Math	8 minutes
Mowing Lawns	20 minutes

The table shows the time it takes Lucinda to earn \$10 while working each of the three listed jobs. If Lucinda cuts hair for 30 minutes, tutors math for 20 minutes, and then mows lawns for 15 minutes, how much money would she have earned?

- \$27.50
- \$40.00
- \$52.50
- \$62.5

46. Answer the question below by choosing the correct response.

$$2 \times 4 + 5 \times 4$$

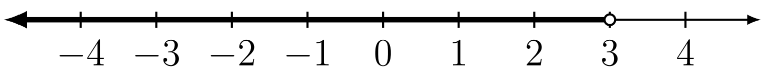
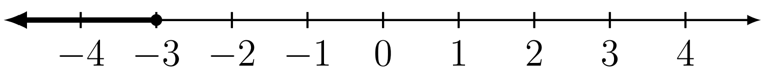
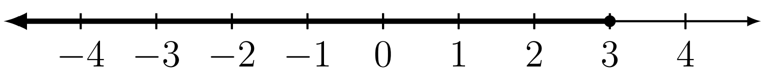
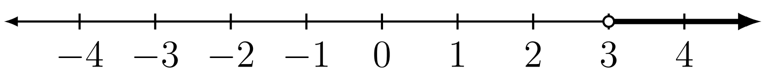
The expression shows the result when the distributive property of multiplication over addition is applied to which of the following expressions?

- ◯ $8 + 20$
- ◯ $(2 + 5) \times 4$
- ◯ $2 + 2 + 2 + 2 + 5 + 5 + 5 + 5$
- ◯ $(5 \times 4) + (2 \times 4)$

47. Answer the question below by choosing the correct response.

$$4 - 2x \geq -\frac{1}{3}x - 1$$

Which of the following is the graph of the solutions to the inequality shown?

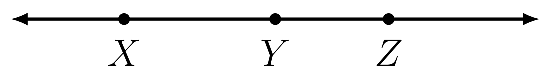
- 
- 
- 
- 

48. Answer the question below by choosing the correct response.

Which of the following list all of the factors of 36

- 2, 3
- 3, 4, 6, 9
- 2, 3, 4, 6, 9, 12, 18
- 1, 2, 3, 4, 6, 9, 12, 18, 36

49. Answer the question below by choosing the correct response.



Which of the following can be used to name the ray that has endpoint Z and goes in the direction of X in the figure shown?

- $\overrightarrow{ZX}$ only
- $\overrightarrow{ZY}$ only
- $\overrightarrow{ZX}$ and $\overrightarrow{ZY}$
- $\overrightarrow{ZX}$ and $\overrightarrow{XZ}$

50. Answer the question below by choosing the correct response.

The function $f\left(x\right) = -x^2 + 1$ is represented by which of the following tables?

x	$f(x)$
-2	-3
0	1
1	0
4	-15

x	$f(x)$
-2	5
0	1
1	0
4	-15

x	$f(x)$
-2	5
0	1
1	2
4	17

x	$f(x)$
-2	5
0	1
1	-1
4	-7